

Program – 9: Organizing Kubernetes Pods Using Namespaces and Labels

Aim

To create and manage Kubernetes pods using namespaces and labels in a Minikube cluster on Ubuntu.

Requirements

- Ubuntu system
- Minikube
- kubectl
- namespaces.yaml, pods.yaml

Step 1: Start Minikube

```
minikube start
```

Step 2: Create Namespaces

namespaces.yaml

```
apiVersion: v1
```

```
kind: Namespace
```

```
metadata:
```

```
  name: dev
```

```
---
```

```
apiVersion: v1
```

```
kind: Namespace
```

```
metadata:
```

```
  name: prod
```

```
kubectl apply -f namespaces.yaml
```

```
kubectl get namespaces
```

Step 3: Create Pods with Labels

pods.yaml

```
apiVersion: v1
```

```
kind: Pod
```

```
metadata:
```

```
  name: frontend-pod
```

```
namespace: dev
labels:
  app: frontend
  env: dev
spec:
  containers:
  - name: frontend
    image: nginx
    ports:
    - containerPort: 80
```

```
apiVersion: v1
kind: Pod
metadata:
  name: backend-pod
  namespace: dev
  labels:
    app: backend
    env: dev
```

```
spec:
  containers:
  - name: backend
    image: nginx
    ports:
    - containerPort: 80
```

```
apiVersion: v1
kind: Pod
metadata:
  name: frontend-pod
  namespace: prod
```

```
labels:
  app: frontend
  env: prod
spec:
  containers:
  - name: frontend
    image: nginx
    ports:
    - containerPort: 80
```

```
apiVersion: v1
kind: Pod
metadata:
  name: backend-pod
  namespace: prod
  labels:
    app: backend
    env: prod
spec:
```

```
  containers:
  - name: backend
    image: nginx
    ports:
    - containerPort: 80
```

kubectI apply -f pods.yamlI

kubectI get pods -n dev

kubectI get pods -n prod

Step 4: Filter Pods Using Labels

kubectI get pods -n dev -I app=frontend

kubectI get pods -n prod -I app=backend

Step 5: View All Pods

kubectl get pods --all-namespaces

```
1rv24mc064_manasas@1rv24mc064manasa: ~/nine
Using Docker driver with root privileges
Starting "minikube" primary control-plane node in "minikube" cluster
Pulling base image v0.0.40 ...
Downloading Kubernetes v1.34.0 preload ...
> preloaded-images-k8s-v18-v1...: 337.07 MiB / 337.07 MiB 100.00% 5.38 Mi
> gcr.io/k8s-minikube/kicbase...: 488.51 MiB / 488.52 MiB 100.00% 1.01 Mi
Creating docker container (CPUs=2, Memory=3072MB) ...
Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...
Configuring bridge CNI (Container Networking Interface) ...
Verifying Kubernetes components...
  ■ Using image gcr.io/k8s-minikube/storage-provisioner:v5
Enabled addons: storage-provisioner, default-storageclass
Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
1rv24mc064_manasas@1rv24mc064manasa:~/nine$ kubectl cluster-info
kubectl get nodes
Kubernetes control plane is running at https://192.168.49.2:8443
CoreDNS is running at https://192.168.49.2:8443/api/v1/namespaces/kube-system/services/kube-dns:proxy

To further debug and diagnose cluster problems, use 'kubectl cluster-info dump'.
NAME      STATUS    ROLES    AGE   VERSION
minikube  Ready     control-plane  15s   v1.34.0
1rv24mc064_manasas@1rv24mc064manasa:~/nine$ cd ~/nine
kubectl apply -f namespaces.yaml
kubectl apply -f pods.yaml
namespace/dev created
namespace/prod created
pod/frontend-pod created
pod/backend-pod created
pod/frontend-pod created
pod/backend-pod created
1rv24mc064_manasas@1rv24mc064manasa:~/nine$ kubectl get pods -n dev
NAME      READY   STATUS    RESTARTS   AGE
backend-pod 0/1     ContainerCreating 0          6s
frontend-pod 0/1     ContainerCreating 0          7s
1rv24mc064_manasas@1rv24mc064manasa:~/nine$ kubectl get pods -n dev -l app=frontend
NAME      READY   STATUS    RESTARTS   AGE
frontend-pod 1/1     Running   0          73s
1rv24mc064_manasas@1rv24mc064manasa:~/nine$ kubectl get pods -n prod -l app=backend
NAME      READY   STATUS    RESTARTS   AGE
backend-pod 1/1     Running   0          79s
1rv24mc064_manasas@1rv24mc064manasa:~/nine$
```

Step 6: Cleanup

kubectl delete pods --all -n dev

kubectl delete pods --all -n prod

kubectl delete namespace dev

kubectl delete namespace prod

Result

Pods were created successfully in separate namespaces and organized using labels.